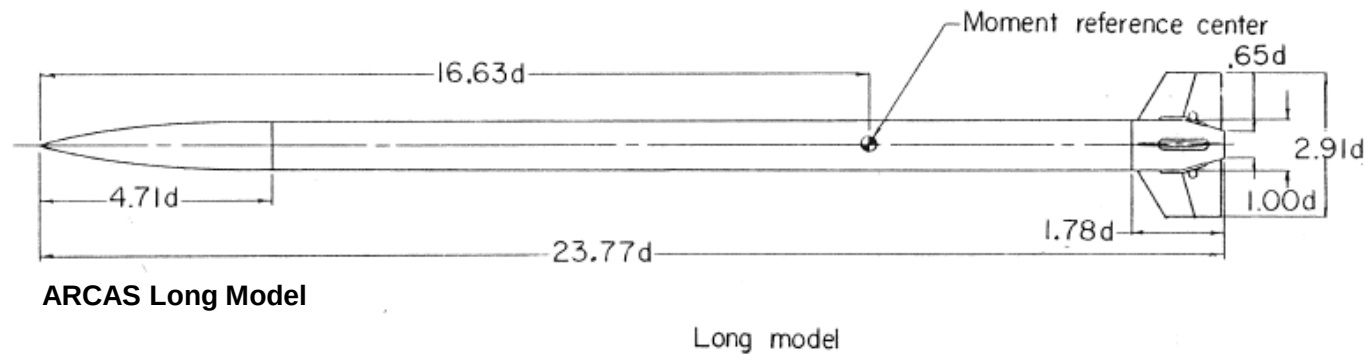
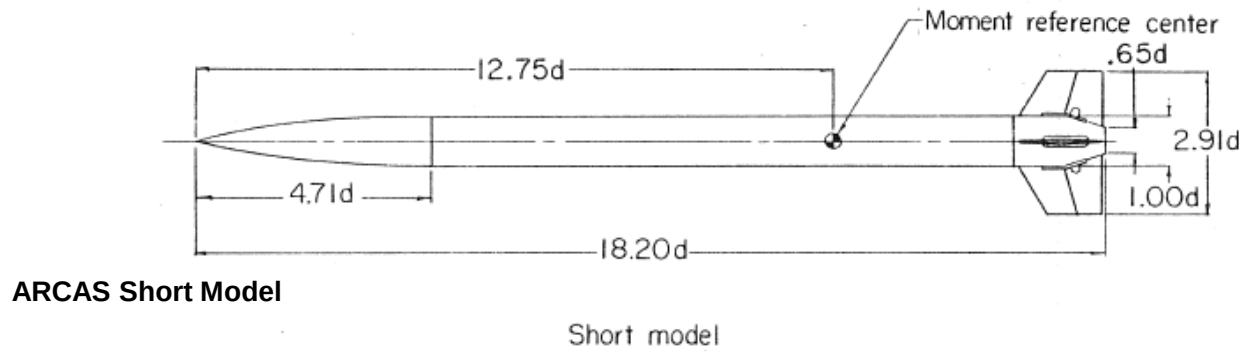


**RASAero II Comparisons with ARCAS
Center of Pressure (CP) and Drag Coefficient (CD)
Wind Tunnel Data**

**Wind Tunnel Data Sources
NASA TN D-4013 and TN D-4014**

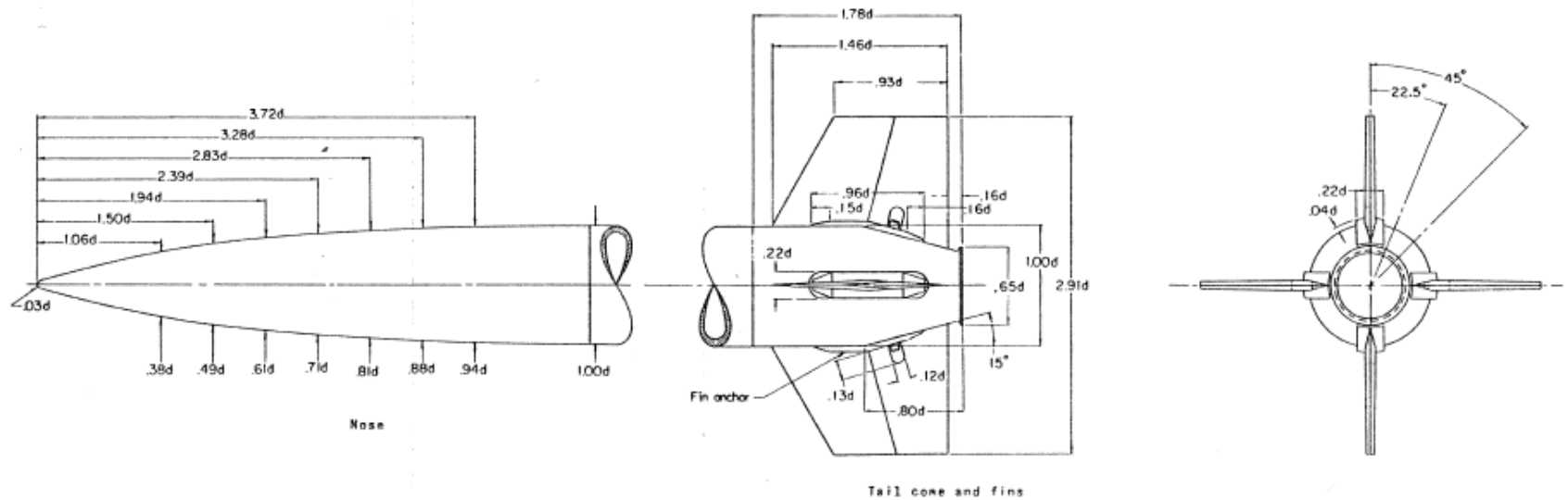
From NASA TN D-4013
ARCAS Sounding Rocket



(a) General arrangement of short and long models.

Figure 1.- Model geometric characteristics. All dimensions are in terms of centerbody diameter d , 5.72 cm (2.25 in.).

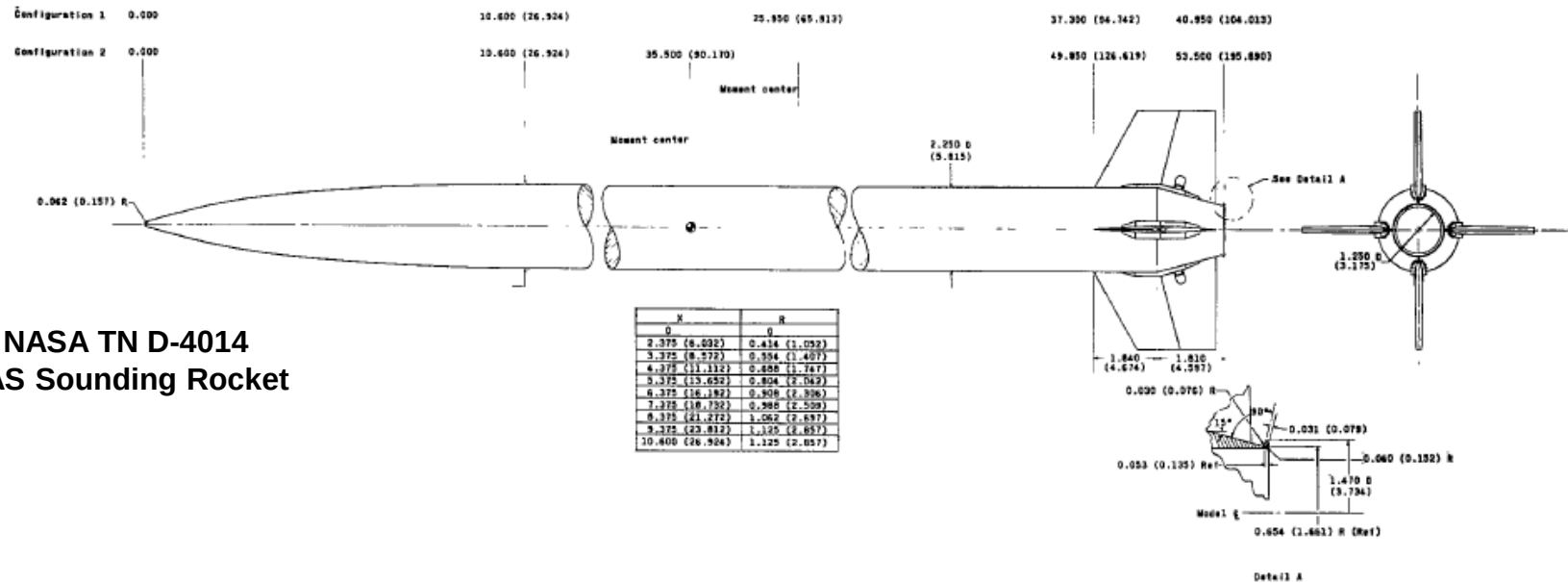
From NASA TN D-4013
ARCAS Sounding Rocket



(b) Details of forebody, tail cone, and fins.

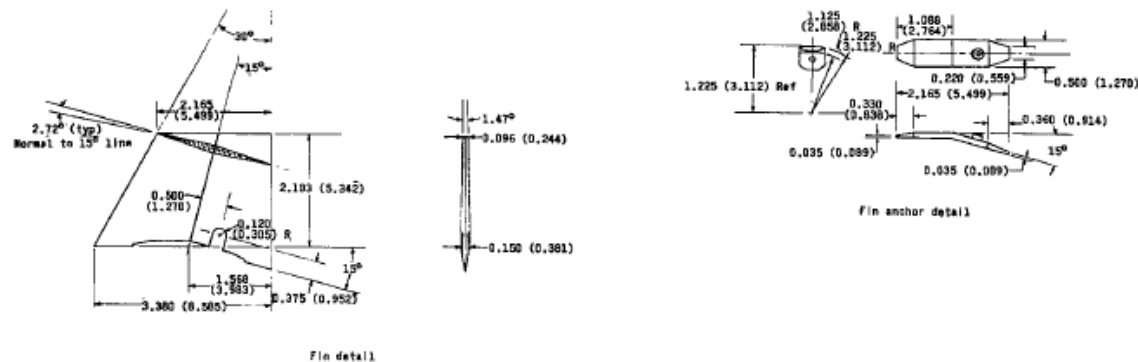
Figure 1.- Concluded.

Configuration 1 is ARCAS Short Model
Configuration 2 is ARCAS Long Model



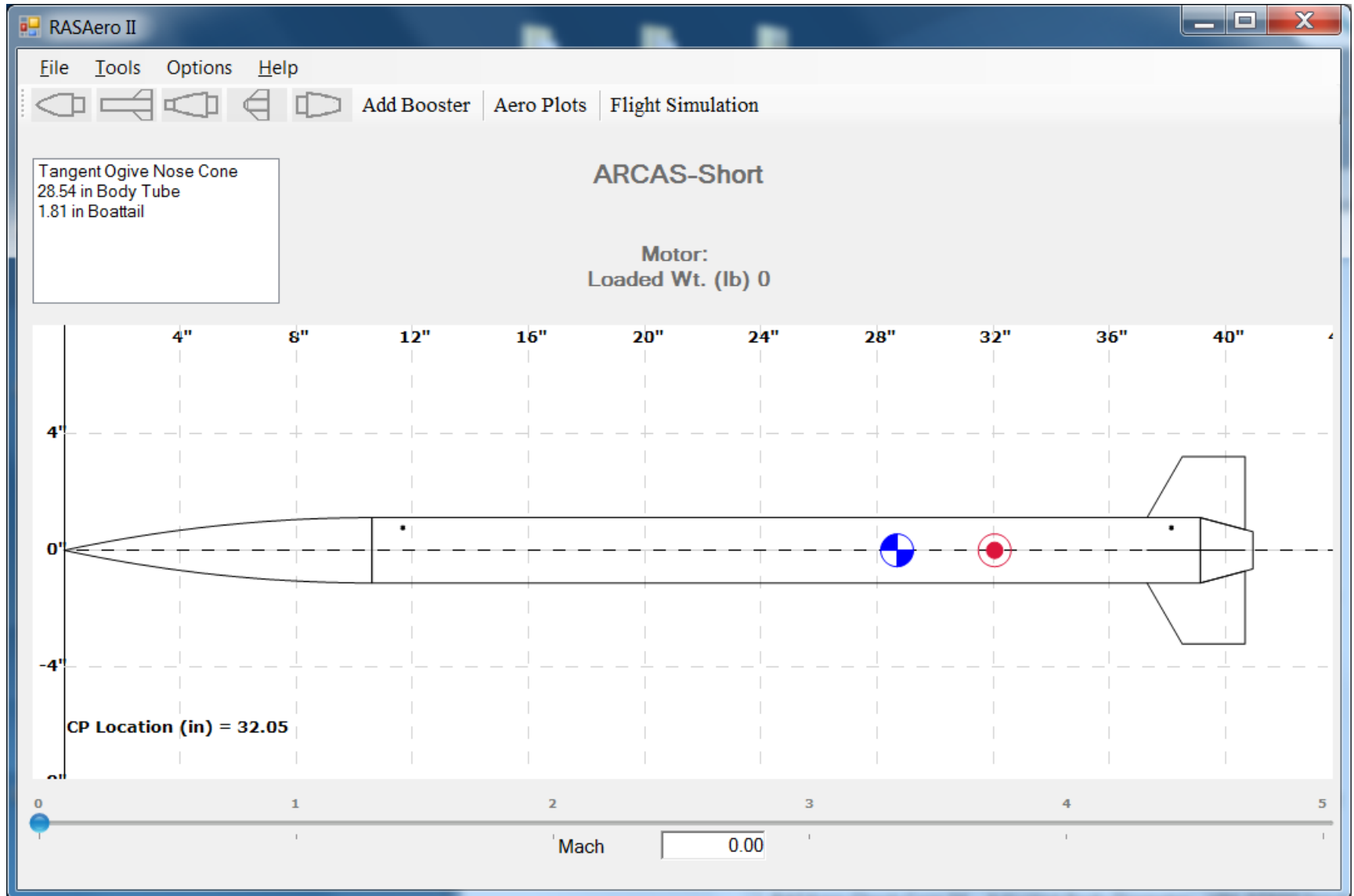
From NASA TN D-4014
ARCAS Sounding Rocket

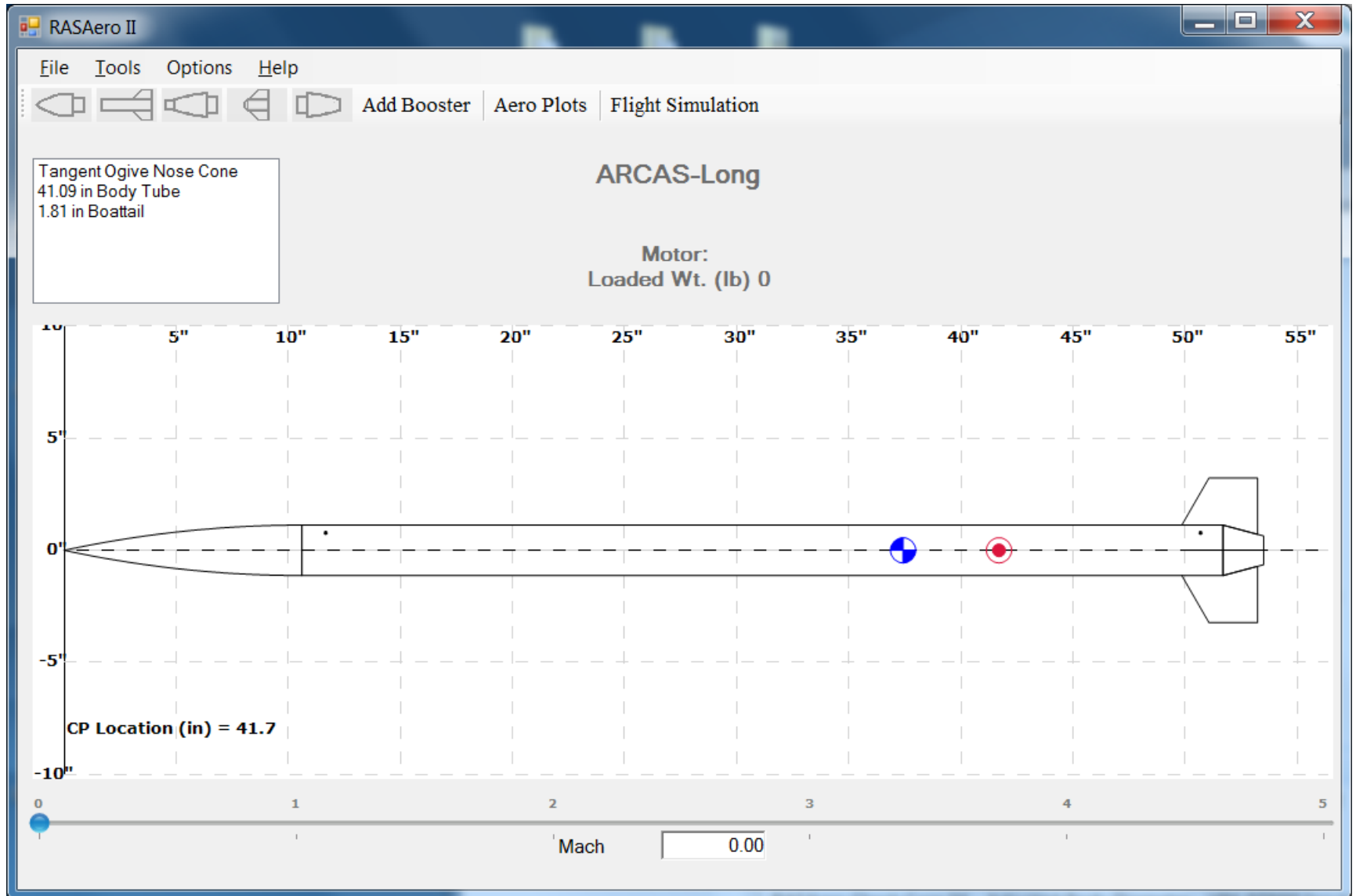
(a) Complete model.



(b) Fin details.

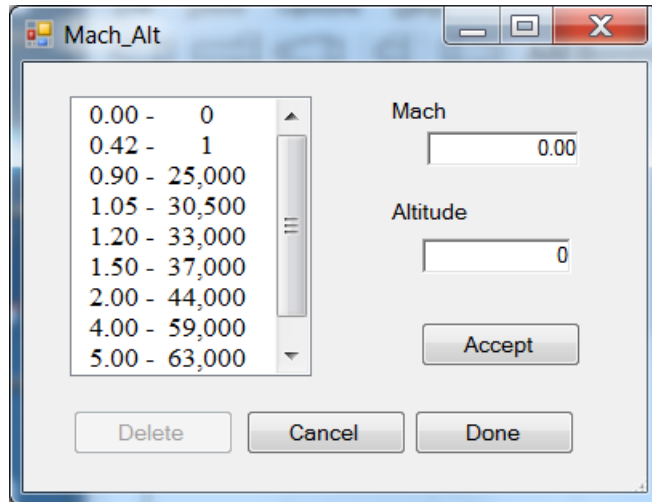
Figure 1.- Model details. All dimensions are in in. (cm).



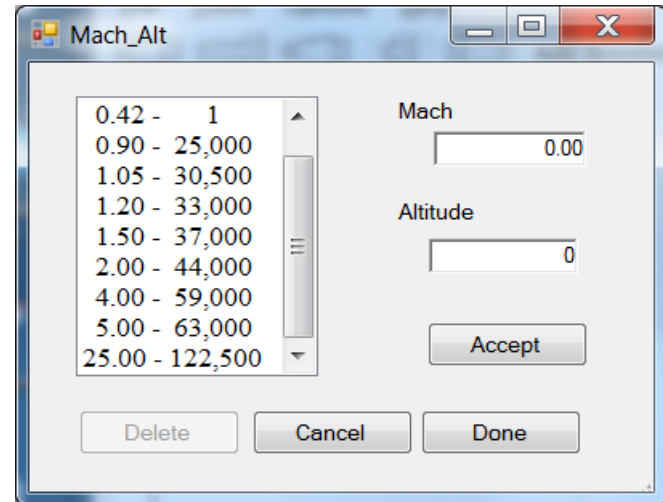


RASAAero II Mach-Alt Inputs for Matching NASA TN D-4013 and TN D-4014 Wind Tunnel Reynolds Number.

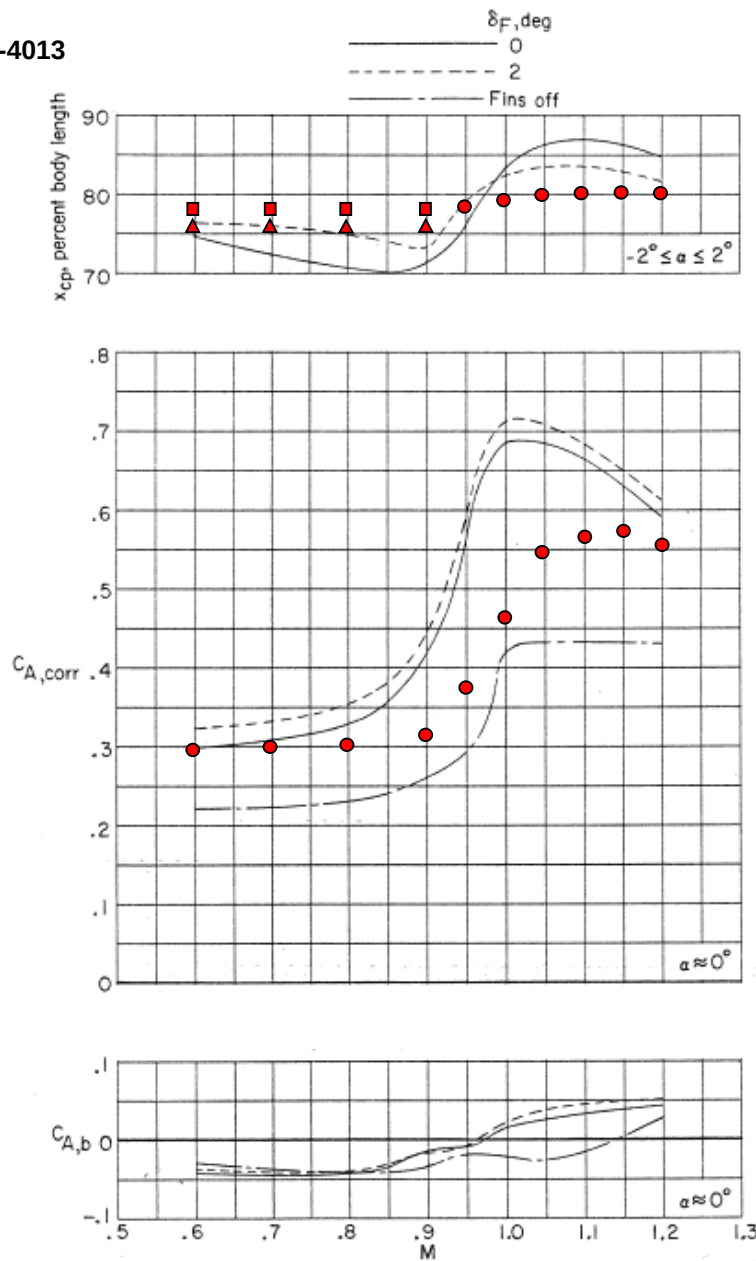
For Both NASA TN D-4013 and TN D-4014 Wind Tunnel Reynolds Number was 3.0×10^6 per foot for Both ARCAS Short and ARCAS Long Configurations.



The screenshot shows the 'Mach_Alt' dialog box. On the left, a list box contains the following Mach-Alt pairs: 0.00 - 0, 0.42 - 1, 0.90 - 25,000, 1.05 - 30,500, 1.20 - 33,000, 1.50 - 37,000, 2.00 - 44,000, 4.00 - 59,000, and 5.00 - 63,000. To the right of the list are two input fields: 'Mach' with a value of 0.00 and 'Altitude' with a value of 0. Below these fields is an 'Accept' button. At the bottom of the dialog are three buttons: 'Delete', 'Cancel', and 'Done'.



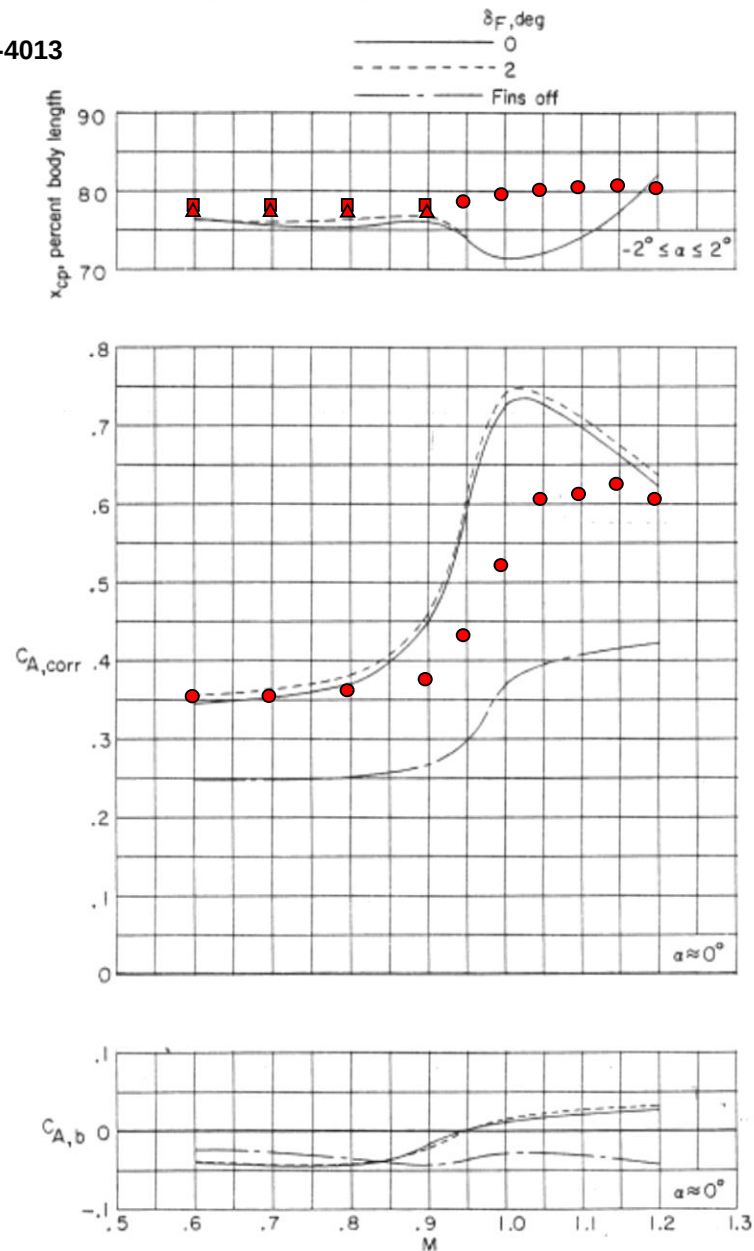
The screenshot shows the 'Mach_Alt' dialog box. On the left, a list box contains the following Mach-Alt pairs: 0.42 - 1, 0.90 - 25,000, 1.05 - 30,500, 1.20 - 33,000, 1.50 - 37,000, 2.00 - 44,000, 4.00 - 59,000, 5.00 - 63,000, and 25.00 - 122,500. To the right of the list are two input fields: 'Mach' with a value of 0.00 and 'Altitude' with a value of 0. Below these fields is an 'Accept' button. At the bottom of the dialog are three buttons: 'Delete', 'Cancel', and 'Done'.



- RASAero II Transonic-Supersonic CP Prediction
- RASAero II Subsonic CP Prediction Rogers Modified Barrowman Method
- ▲ RASAero II Implementation of Barrowman Method

- RASAero II CD Prediction (Power-Off)

Figure 11.- Variation of center-of-pressure location, $C_{A,corr}$ and $C_{A,b}$ with Mach number. Short model; $\phi = 0^\circ$, (Roll Angle 0 deg)



- RASAero II Transonic-Supersonic CP Prediction
- RASAero II Subsonic CP Prediction Rogers Modified Barrowman Method
- ▲ RASAero II Implementation of Barrowman Method

- RASAero II CD Prediction (Power-Off)

Figure 12.- Variation of center-of-pressure location, $C_{A,corr}$ and $C_{A,b}$ with Mach number. Long model; $\Phi = 0^\circ$, (Roll Angle 0 deg)

NASA TN D-4014

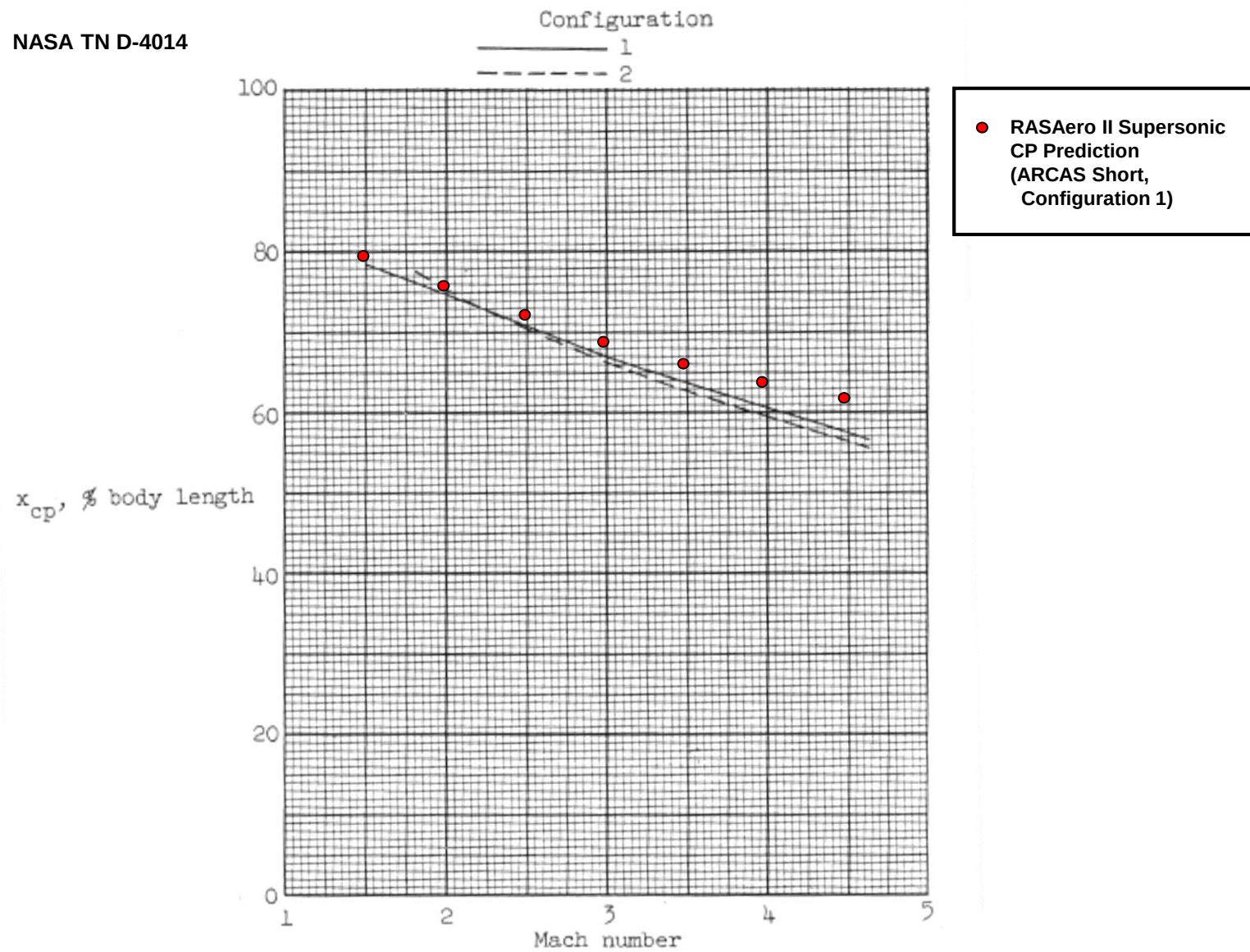


Figure 7.- Variation of center of pressure with Mach number at low angles of attack.

NASA TN D-4014

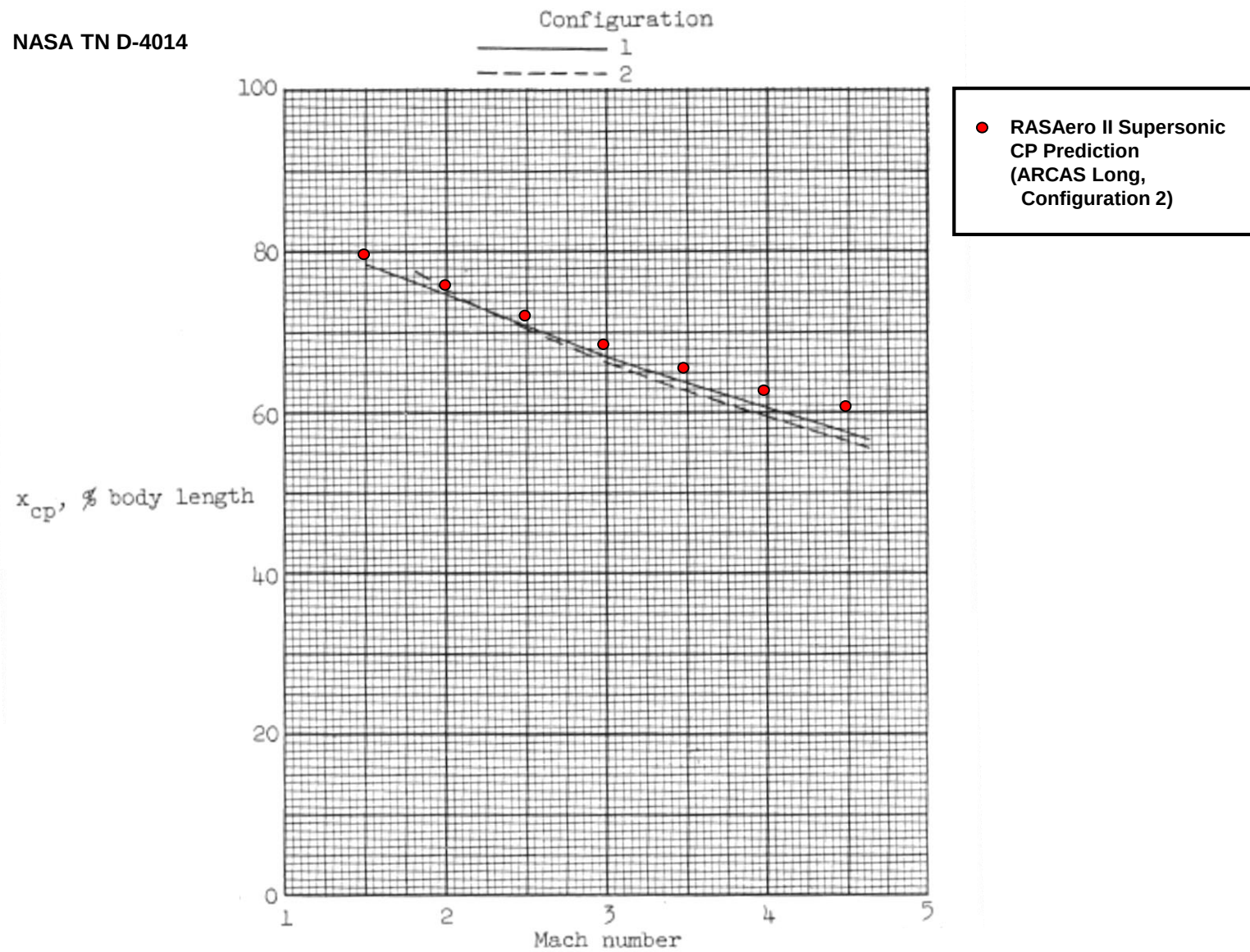


Figure 7.- Variation of center of pressure with Mach number at low angles of attack.

NASA TN D-4014

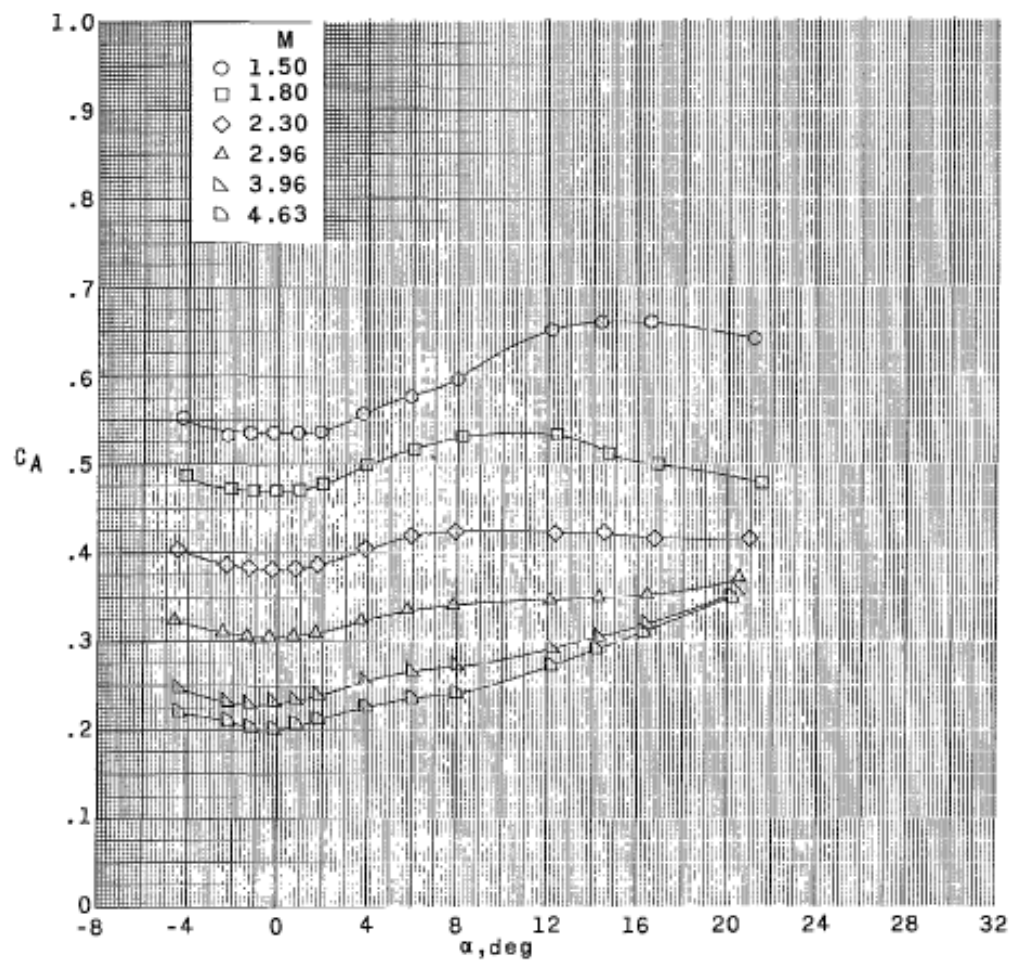
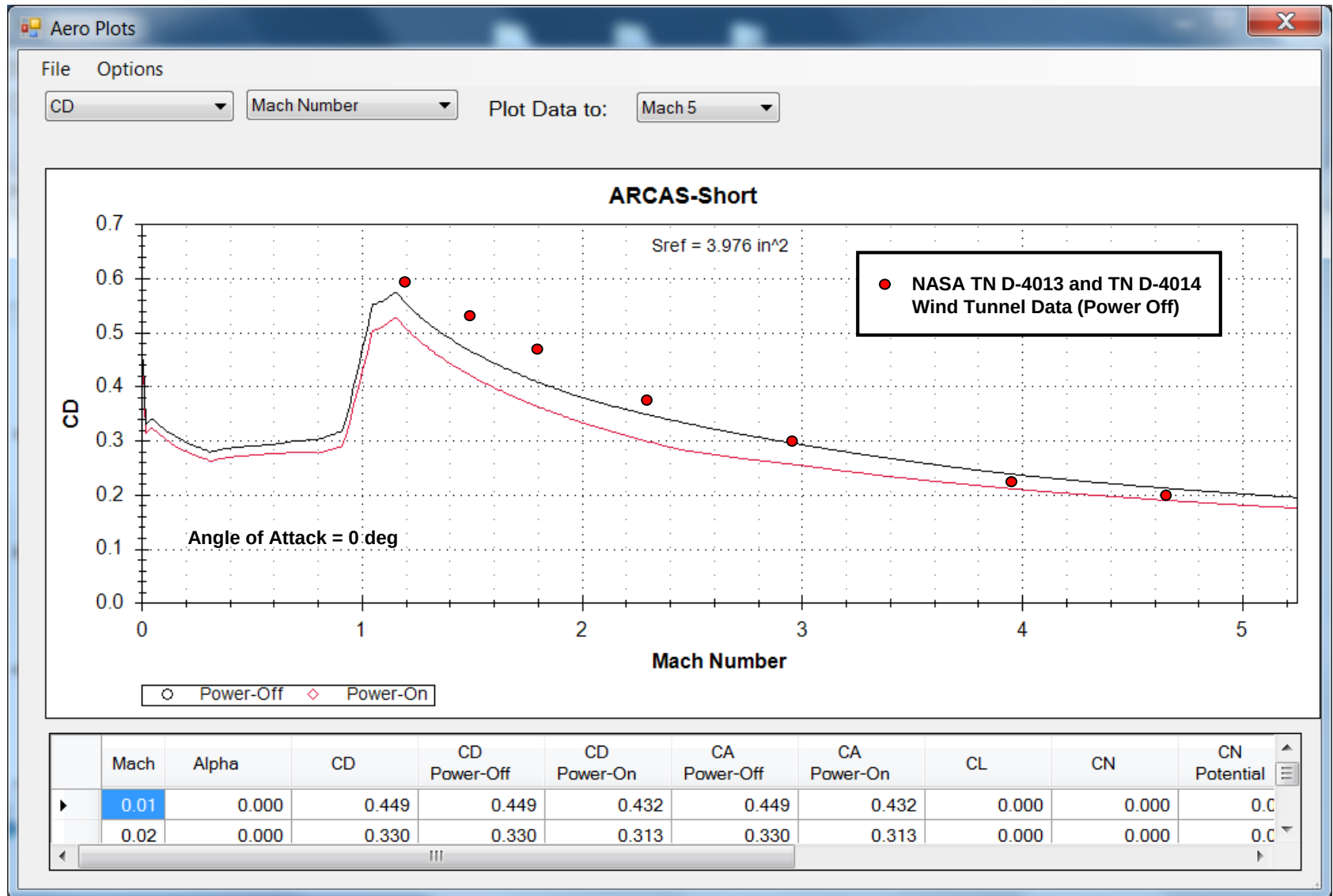
(b) $\delta_F = 0^\circ$.

Figure 5.- Continued.

Figure 5.- Aerodynamic characteristics in pitch for configuration 1. (ARCAS Short)



NASA TN D-4014

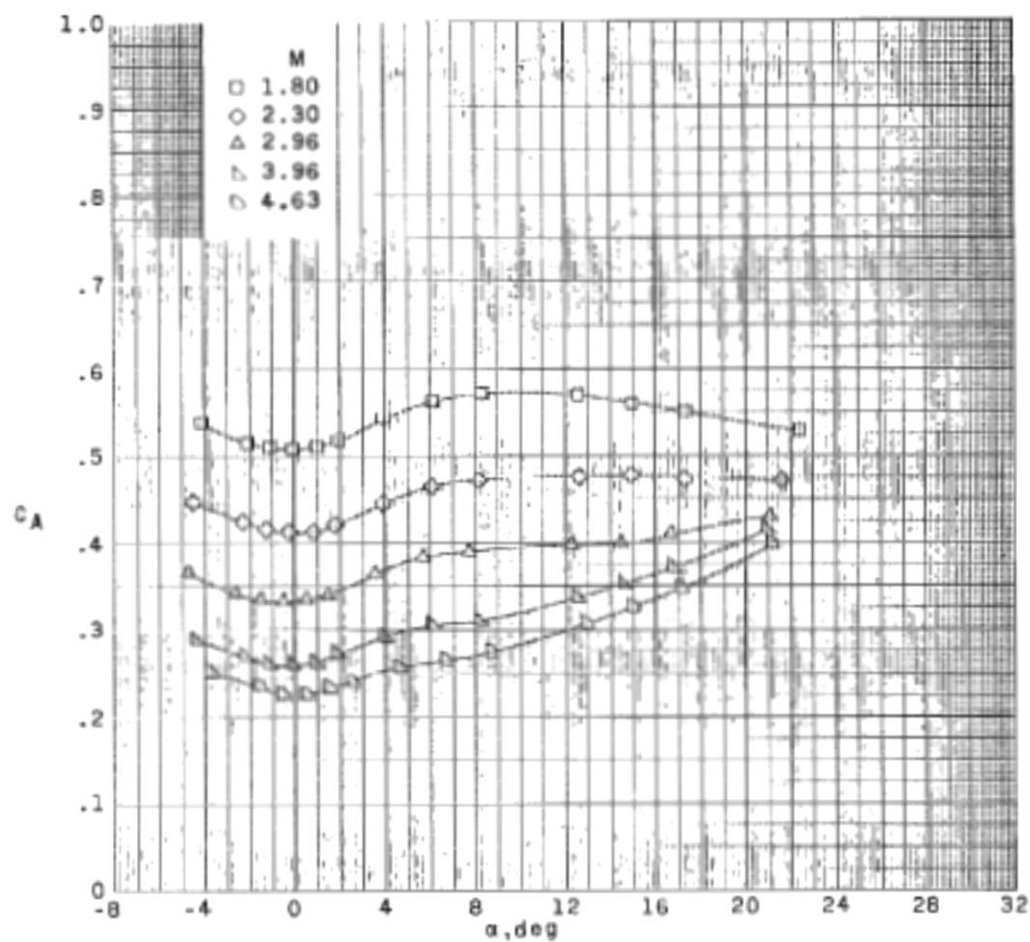
(b) $\delta_F = 0^\circ$.

Figure 6.- Continued.

Figure 6.- Aerodynamic characteristics in pitch for configuration 2. (ARCAS Long)

